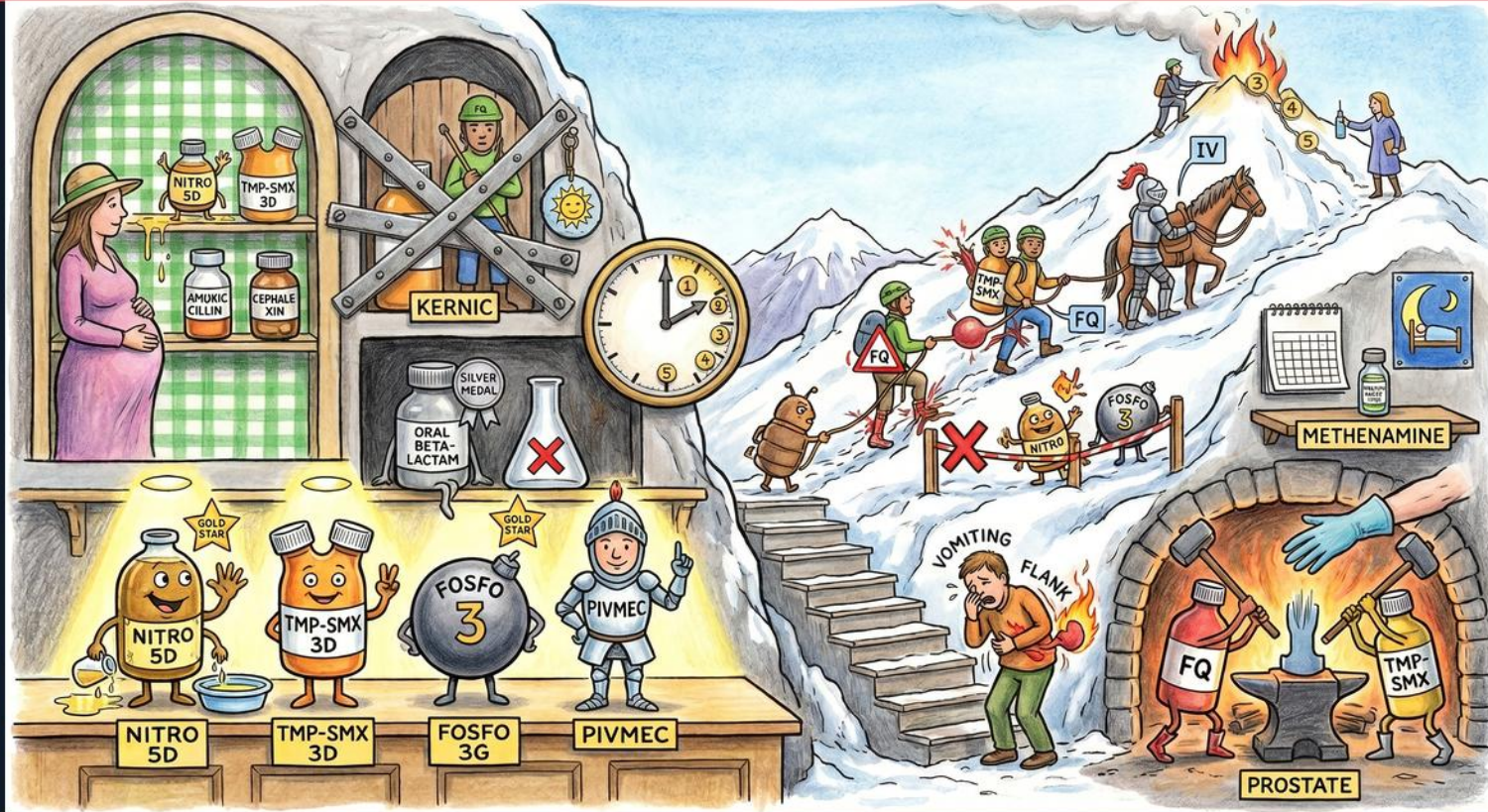
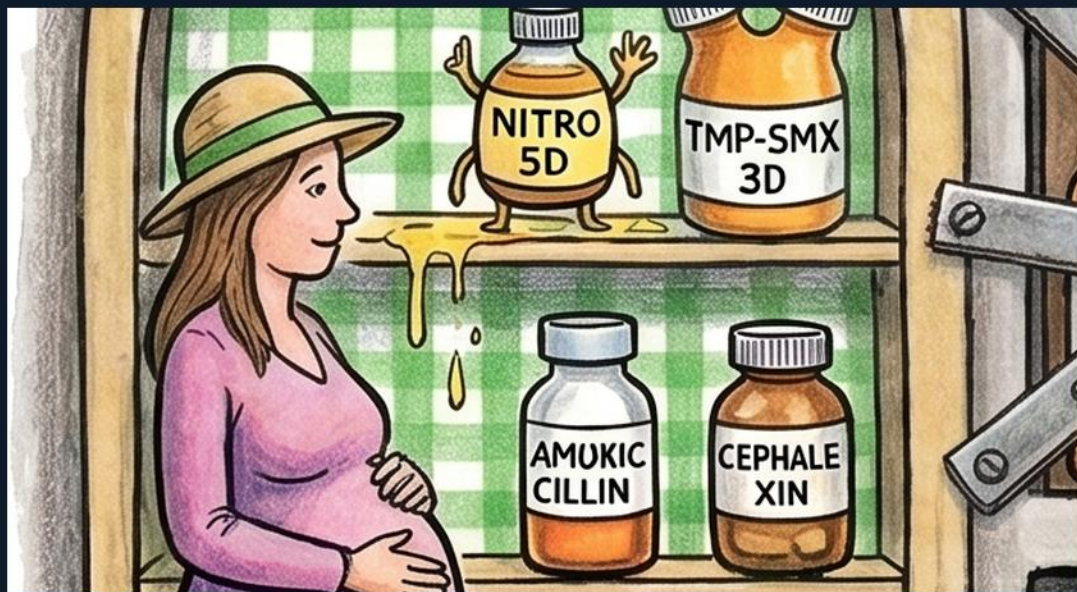


UTI — Treatment & Special Populations





Pregnancy-safe window

WHAT YOU SEE

Pregnant woman at the green-checkered apothecary window beside 'NITRO 5D', 'AMOXICILLIN', 'CEPHALEXIN' jars — the pregnancy-safe shelf

WHAT IT ENCODES

Pregnancy-SAFE agents: nitrofurantoin (avoid near term), amoxicillin/amoxicillin-clavulanate, cephalexin and other cephalosporins, and fosfomycin. Screen all pregnant women for asymptomatic bacteriuria and treat positives 4-7 days with a test-of-cure.



KERNIC crossed-out (avoid near term)

WHAT YOU SEE

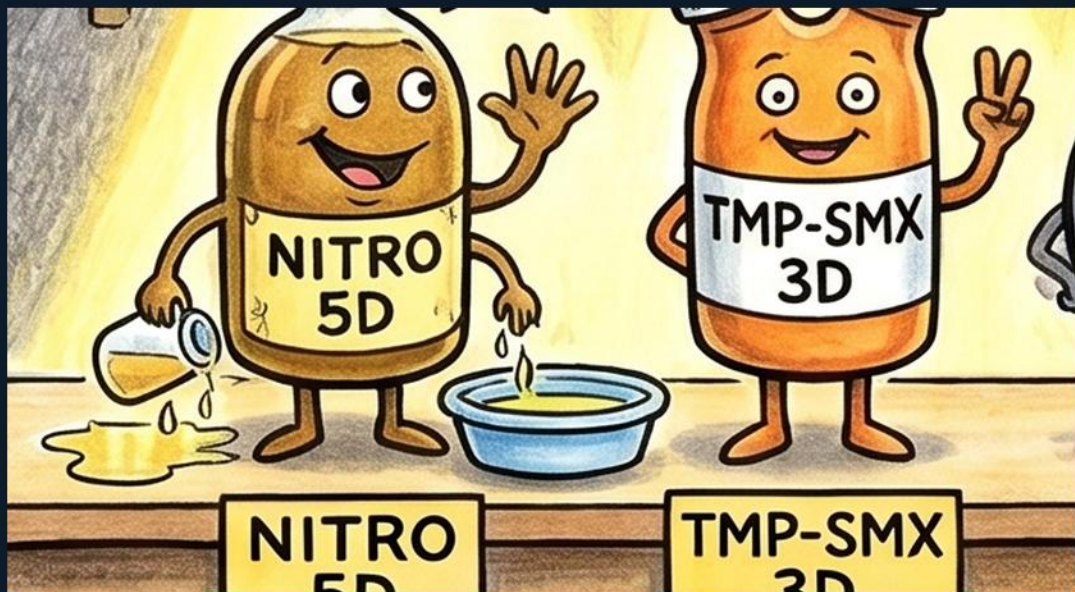
Soldier behind a big black X labeled 'KERNIC' with a setting sun — drugs to AVOID near term/3rd trimester

WHAT IT ENCODES

Near term / 3rd trimester AVOID sulfonamides (TMP-SMX) and nitrofurantoin: risk of neonatal kernicterus (sulfa displaces bilirubin) and hemolytic anemia (especially with G6PD deficiency). Throughout pregnancy AVOID fluoroquinolones (fetal cartilage) and TMP-SMX in the 1st trimester (folate antagonist -> neural tube defects).

THE BOARD TRAP

"TMP-SMX or nitrofurantoin near delivery" -> wrong; sulfa near term -> kernicterus, nitrofurantoin near term -> neonatal hemolysis — pick a cephalosporin or amoxicillin instead.



Nitrofurantoin 5d (1st-line, gold star)

WHAT YOU SEE

GOLD-STAR potion bottle labeled 'NITRO 5D' on the front shelf — a first-line cystitis champion

WHAT IT ENCODES

Nitrofurantoin 100 mg BID x5 days is first-line for uncomplicated cystitis — it concentrates in URINE only, keeps excellent E. coli activity with low collateral resistance, and is generally pregnancy-safe (except near term).

THE BOARD TRAP

"nitrofurantoin for pyelonephritis/prostatitis" -> wrong; no tissue/parenchymal penetration. Also ineffective when CrCl <30 (weak 30-45) and avoid near term and in G6PD deficiency (hemolysis).



TMP-SMX 3d (1st-line if <20%)

WHAT YOU SEE

Bottle labeled 'TMP-SMX 3D' on the front shelf — first-line cystitis where resistance is low

WHAT IT ENCODES

TMP-SMX 1 DS tablet BID x3 days is first-line for uncomplicated cystitis WHERE local *E. coli* resistance is <20%. It penetrates tissue, so it also serves in pyelonephritis (10-14 d) and prostatitis.

THE BOARD TRAP

"empiric TMP-SMX when resistance >20% or used in last 3 months" -> wrong; pick another agent. Avoid 1st trimester (NTDs) and near term (kernicterus); watch hyperkalemia and warfarin interaction.



Fosfomycin 3g (gold-medal single dose)

WHAT YOU SEE

GOLD-MEDAL cannonball bottle labeled 'FOSFO 3' on the shelf — a single-dose cystitis winner

WHAT IT ENCODES

Fosfomycin is a single 3 g oral dose first-line cystitis option — convenient and active against many resistant organisms including ESBL-producing *E. coli* and VRE. It concentrates in urine.

THE BOARD TRAP

"fosfomycin for febrile/upper-tract UTI" -> wrong; single-dose fosfomycin is CYSTITIS only and does not reach tissue levels for pyelonephritis or prostatitis.



Pivmecillinam (1st-line cystitis)

WHAT YOU SEE

Knight bottle labeled 'PIVMEC' on the shelf — a beta-lactam first-line cystitis agent

WHAT IT ENCODES

Pivmecillinam (a beta-lactam prodrug of mecillinam) is a first-line cystitis agent in Europe/Israel with low resistance and a good safety profile including pregnancy. Like the other cystitis drugs it treats lower-tract disease only.

THE BOARD TRAP

"pivmecillinam for pyelonephritis" -> wrong; reserve it for cystitis.



Oral beta-lactam (2nd-line, silver + red-X)

WHAT YOU SEE

'ORAL BETA LACTAM' jar with a SILVER MEDAL and a red-X flask — runner-up, less-effective cystitis option

WHAT IT ENCODES

Oral beta-lactams (amoxicillin-clavulanate, cefpodoxime, cephalexin) are SECOND-line for cystitis — lower cure rates and more relapse than first-line agents, partly from incomplete eradication of vaginal/gut reservoirs. Use when first-line drugs can't be used.

THE BOARD TRAP

"plain amoxicillin/ampicillin empirically for cystitis" -> wrong; high E. coli resistance makes bare aminopenicillins a poor empiric choice.



Fluoroquinolone (red-X, reserved)

WHAT YOU SEE

Red-X bee/figure labeled 'FQ' roped off on the mountain — fluoroquinolones blocked from first-line cystitis

WHAT IT ENCODES

Fluoroquinolones (ciprofloxacin, levofloxacin) are RESERVED for pyelonephritis and complicated UTI/prostatitis where tissue penetration matters. FDA boxed warnings — tendinopathy/rupture, aortic aneurysm/dissection, irreversible peripheral neuropathy, CNS effects, QT prolongation, C. difficile — mean avoid when safer options exist.

THE BOARD TRAP

"fluoroquinolone first-line for simple cystitis" -> wrong; that is collateral damage. Also avoid in pregnancy and children (cartilage toxicity).



Fosfomycin red-X on mountain

WHAT YOU SEE

Cannonball labeled 'FOSFO 3' behind a red X on the mountain climb — fosfomycin barred from the upper tract

WHAT IT ENCODES

The mountain = pyelonephritis (upper tract). Urine-only agents are explicitly OFF the table here: neither fosfomycin nor nitrofurantoin reach parenchymal levels. Upper-tract disease demands fluoroquinolones, tissue-penetrating beta-lactams, or TMP-SMX.

THE BOARD TRAP

"fosfomycin (or nitrofurantoin) for pyelonephritis" -> wrong; 'no tissue penetration' is the trap — choose a drug with renal parenchymal levels.

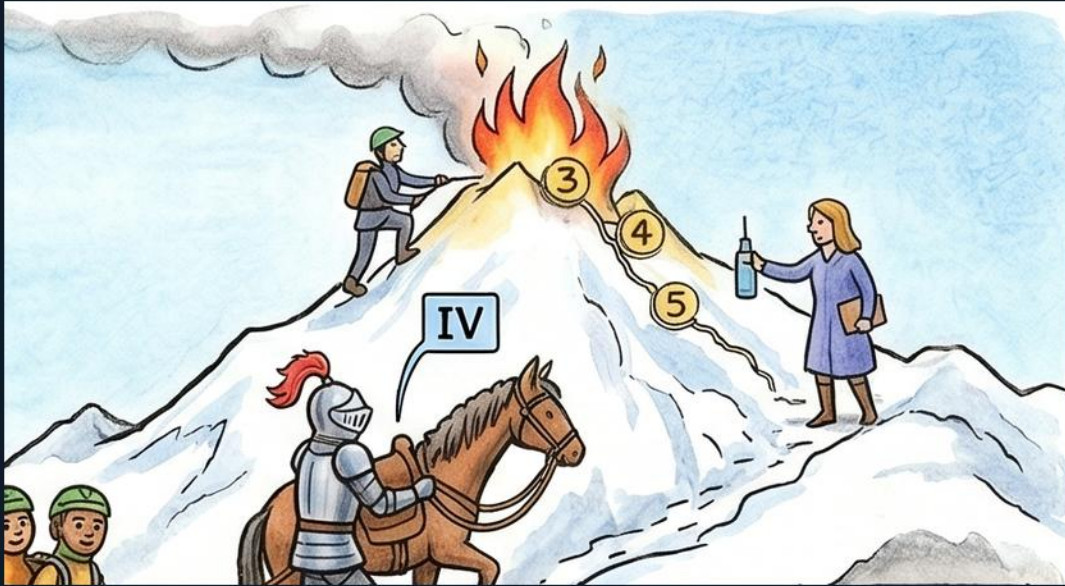
IV climb (inpatient pyelo)

WHAT YOU SEE

'IV' climbers and knight on horseback ascending the snowy peak — inpatient pyelonephritis going up the mountain

WHAT IT ENCODES

Severe/inpatient pyelonephritis is treated IV: ceftriaxone or another broad beta-lactam (pip-tazo, cefepime, or a carbapenem if ESBL risk), plus an aminoglycoside for critically ill/resistant cases. De-escalate to oral once afebrile and culture-guided. Duration: pyelonephritis 5-7 d with a fluoroquinolone, 7 d for IV beta-lactam, 10-14 d for TMP-SMX/oral beta-lactam.





Vomiting / flank (admission criteria)

WHAT YOU SEE

Figure on the staircase clutching a burning flank labeled 'VOMITING / FLANK' — the patient who must be admitted

WHAT IT ENCODES

Admission criteria for pyelonephritis: inability to tolerate PO (vomiting), sepsis/instability, pregnancy, immunocompromise, suspected obstruction, or outpatient failure. No improvement within 48-72 h -> image with CT/ultrasound to exclude obstructing stone, perinephric abscess, or emphysematous change.

THE BOARD TRAP

"keep the same regimen past 72 h of fever" -> wrong; persistent fever on appropriate antibiotics means image for a complication (abscess/obstruction).



Methenamine (recurrent prophylaxis)

WHAT YOU SEE

Moon/calendar panel labeled 'METHENAMINE' on the right cliff — non-antibiotic prophylaxis for recurrent UTI

WHAT IT ENCODES

Recurrent UTI = ≥ 2 in 6 months or ≥ 3 in 12 months.

Options: continuous low-dose prophylaxis (nitrofurantoin or TMP-SMX) for 6+ months, post-coital prophylaxis if intercourse-related, and patient-initiated self-treatment.

Non-antibiotic: methenamine (urinary antiseptic), topical vaginal estrogen in postmenopausal women (restores lactobacilli/lowers pH), and behavioral measures.

THE BOARD TRAP

"cranberry / lifelong antibiotics first" -> wrong; evaluate for a correctable structural cause before long-term antibiotics, and cranberry has weak/inconsistent evidence.



Prostate forge (men)

WHAT YOU SEE

Blacksmith forge labeled 'PROSTATE' with 'FQ' and 'TMP-SMX' tools — UTI in men means think prostatitis

WHAT IT ENCODES

UTI in a man is by definition complicated — always consider PROSTATITIS as the source and pick prostate-penetrating agents: fluoroquinolones or TMP-SMX. Acute bacterial prostatitis 2-4 weeks; chronic bacterial prostatitis 4-6 weeks because of poor penetration of the non-inflamed gland.

THE BOARD TRAP

"prostate massage in acute prostatitis" -> wrong; risks bacteremia. Don't use nitrofurantoin/fosfomycin (no prostate penetration), and a febrile UTI in a man warrants a prostatic/obstructive evaluation.

Summary — every symbol

1. Pregnancy-safe window

Pregnancy-SAFE agents: nitrofurantoin (avoid near term), amoxicillin/amoxicillin-clavulanate, cephalexin and other cephalosporins, and fosfomycin

3. Nitrofurantoin 5d (1st-line, gold star)

Nitrofurantoin 100 mg BID x5 days is first-line for uncomplicated cystitis — it concentrates in URINE only, keeps excellent E

5. Fosfomycin 3g (gold-medal single dose)

Fosfomycin is a single 3 g oral dose first-line cystitis option — convenient and active against many resistant organisms including ESBL-producing E

7. Oral beta-lactam (2nd-line, silver + red-X)

Oral beta-lactams (amoxicillin-clavulanate, cefpodoxime, cephalexin) are SECOND-line for cystitis — lower cure rates and more relapse than first-line agents, partly from incomplete eradication of vaginal/gut reservoirs

9. Fosfomycin red-X on mountain

The mountain = pyelonephritis (upper tract)

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Admission criteria for pyelonephritis: inability to tolerate PO (vomiting), sepsis/instability, pregnancy, immunocompromise, suspected obstruction, or outpatient failure

13. Prostate forge (men)

UTI in a man is by definition complicated — always consider PROSTATITIS as the source and pick prostate-penetrating agents: fluoroquinolones or TMP-SMX

2. KERNIC crossed-out (avoid near term)

Near term / 3rd trimester AVOID sulfonamides (TMP-SMX) and nitrofurantoin: risk of neonatal kernicterus (sulfa displaces bilirubin) and hemolytic anemia (especially with G6PD deficiency)

4. TMP-SMX 3d (1st-line if <20%)

TMP-SMX 1 DS tablet BID x3 days is first-line for uncomplicated cystitis WHERE local E

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Pivmecillinam (a beta-lactam prodrug of mecillinam) is a first-line cystitis agent in Europe/Israel with low resistance and a good safety profile including pregnancy

8. Fluoroquinolone (red-X, reserved)

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10. IV climb (inpatient pyelo)

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